Twins weather and sensor network — a narrative

Updated October 2026. The first version of this was written in July 2026; the camera section in particular has changed since then.

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1. Background — why, where, and the first cameras

I've been building hardware and writing code since I was a teenager — RadioShack parts, TRS-80 in 1979, first paid program at twelve, then cryptography, telephony, sensor frameworks, and more Arduino projects than I can count. I grew up in Maine skiing in the 80s and snowboarding by the late 80s. I like connected gadgets and I like being outside; this project is where those two things collided.

When I moved here four or five years ago, I started exploring the Twins — North Twin, South Twin, Bowman Mountain, Dailey Prairie, and the ridges and drainages between them. I got a permit and kept driving out just to **see** what conditions were like. The Twins gave me a reason to keep showing up in a new place; first trips stuck with me for the terrain, the access logistics (weather windows, snow, gates), and the recurring wish for **eyes on conditions** without driving to the middle of nowhere every time.

Bowman Ridge is an informal name I use — Bowman Mountain has been labeled differently on maps over the years. The ridge runs from Bowman along the west side of the "bathtub" that is Dailey Prairie, toward Blue Mountain Road, and forms the Orsino Creek drainage toward Skookum. It's a strong sensor site: high enough to see Baker and the Twins, often with cell coverage facing town, though the highest points sit around 4,000 feet and stay buried in snow much of the year.

Valley forecasts and airport weather often don't match what's happening on the Twins. Ridge microclimate — wind, temperature, snow cover, visibility — can be completely different from town or a consumer app. That gap is why local observations matter for backcountry skiing, route planning, and go/no-go calls in sketchy weather. For instance, the weather on the SW side of Bowman Ridge is very different from the NE side.

After enough repeat drives, the question was obvious: *why am I doing this when I could put cameras and sensors up there?* So I bought a trail camera, put it on Bowman Mountain, and started watching. Then it got out of control — in a good way. Fire season pushed me toward **air quality on the ridges**, not just in town. Temperature and pressure came first, then snow depth, wind, soil probes, doors, and a LoRaWAN mesh to tie it together.

My friend Anders — who taught me to snowboard in the 80s — jokes that I combined woods time and harsh-environment electronics. Anders (electrical/computer engineer), Gabe, and Dave have helped with battery sizing, equations, and the electrical details I'm still learning. I carry a Starlink kit in the field without cell service — which is how part of this got written while driving out to un-brick a device at sunset, looking at North Twin and Baker with the dogs.

Trail cameras

There are **five cameras** on the project website right now, with a sixth being installed — easier to spot than the small sensor nodes because they need a clear view. I installed the first ones **in the middle of winter**, including trips in freezing rain that were miserable. Placement targets terrain and conditions, not people. They have caught a few passersby anyway, and I'm moving cameras higher in trees and farther from trails where I can still get useful shots without being creepy.

Camera on the site	What it is	Where
Bowman Mtn	Vosker cellular trail cam	~3,200 ft, facing Baker and the Twins
N Twin	GardePro cellular trail cam	~4,200 ft, on the logging road up to the NW ridge
W Ridge N Twin	GardePro cellular trail cam	~4,600 ft, looking at the N Twin skin track
Westward View	GardePro cellular trail cam	~3,900 ft on Bowman Ridge, looking west toward the valley
Bowman Ridge	**Custom camera,** `woods-cam-003` (Raspberry Pi 5)	~4,000 ft, looking at the Twins and Baker above the Middle Fork borrow pit
Bowman Ridge High Point	Reconyx cellular trail cam — being installed, not on the page yet	Bowman Ridge

The camera story started with off-the-shelf **cellular trail cams**, not custom hardware. The Vosker and GardePro units are battery-powered, LTE-connected, and built for remote woods deployment. They wake on a schedule or motion trigger and upload photos to each vendor's cloud app. With help from Claude, I wrote downloaders that pull new images from those services onto the project website — thousands of frames so far. **That path still feeds four of the five cameras on the site.**

The **custom camera** is the one I've spent the most time on. The first version was built on a **Particle Tachyon** single-board computer with a 4K USB camera in a 3D-printed enclosure. It ran on Bowman Ridge through the summer of 2026 as `woods-cam-001`, and it taught me most of what the current version does differently. It was retired to the bench on 2026-10-12. The camera on the ridge today, `woods-cam-003`, is a **Raspberry Pi 5** build:

- a **Witty Pi 5** power-management board owns the power: it turns the Pi on, the Pi does its work and halts, and the board cuts the rail and starts the countdown to the next wake;
- a **Quectel LTE modem** on a Sixfab HAT for the uplink;
- a 4K USB camera (the same Novatek-based module the Tachyon used), with lens correction and a caption burned into every frame on the box;
- a BME280 for temperature, humidity and pressure, so the camera doubles as a weather node;
- all of it in a Particle M1 enclosure on solar and a battery.

Each wake is about three seconds of real work. The box reads the GPS, the clock and the sensor first, so they can be written into the picture, then takes one frame, judges whether it is worth sending (black, blown-out and corrupt frames are kept locally but not uploaded), and only then touches the network: the

frame goes to **Backblaze B2**, and the weather reading and a heartbeat go to the same database as every other sensor. A second Pi 5 build, `woods-cam-002`, is being rebuilt into a new enclosure and is not in the field right now.

Particle also gave me a satellite module I haven't deployed — uplink is expensive — but it would be useful on the far side of the Twins, maybe near the glaciers, where there's no cell.

Diagram — how pictures get to the website

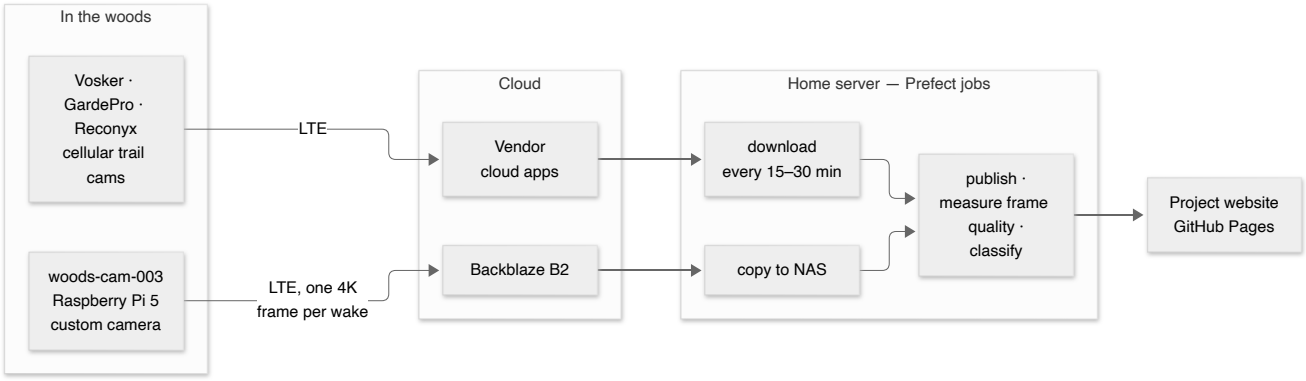
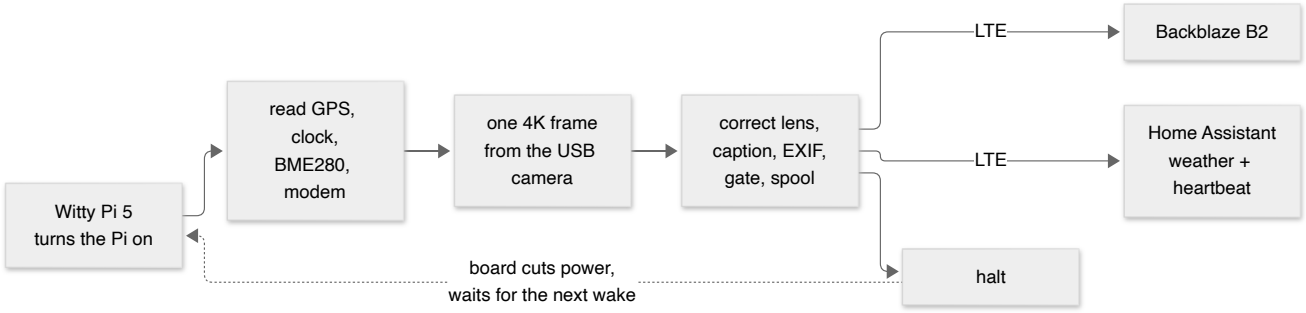


Diagram — one wake of the custom camera



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2. Particle.io fleet

The first custom field sensor was a **Particle Boron** — a small cellular microcontroller board — on Bowman. It worked, and it didn't: cellular up there is sketchy. That pushed two paths at once — **cellular** where coverage exists, **LoRaWAN** where it doesn't (or as backup). Two Borons exist today: one in the field on the 7000 Road, one on the bench.

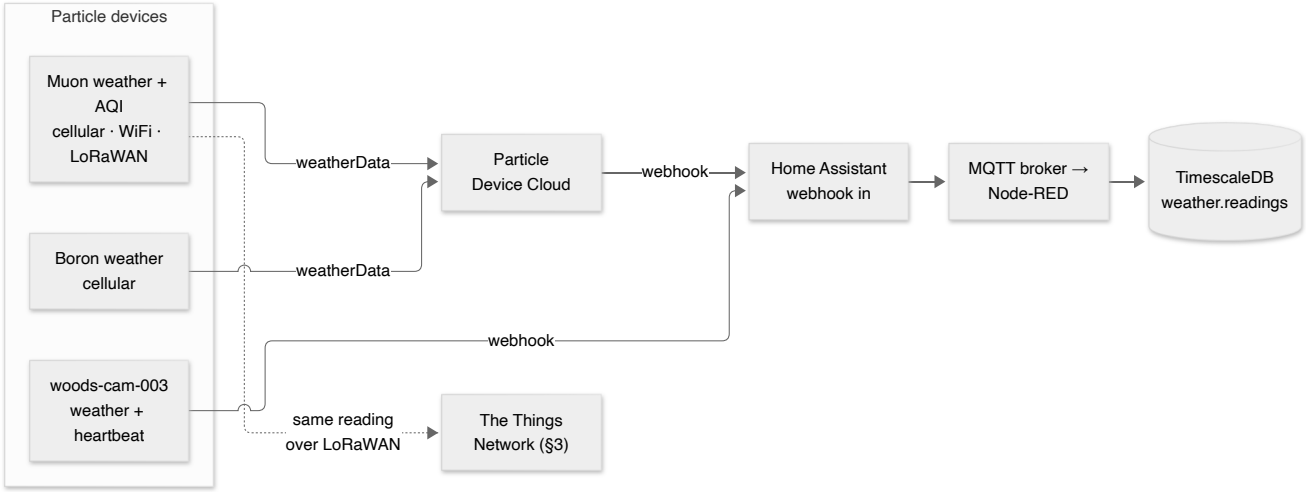
I've since built **six custom weather and air-quality stations** on **Particle Muon** hardware (Wi-Fi, Ethernet, LoRaWAN, cellular — deploy with whatever fits the site). They use Sensirion SEN55 particulate sensors over I2C. The SEN55 shares an I2C address with a clock chip on the Muon board, which cost me a wrong turn through a UART bridge before an address translator fixed it properly. The newest build adds an **ultrasonic wind sensor** and a **solar battery pack that reports its own voltage and current**

over RS-485, so the station can tell me how its power is doing. They're more noticeable than Dragino nodes and expensive — roughly $500–600 each in parts, not counting hours. I built them as a labor of love during fire season; most parts were on hand, and an LLC absorbs some project costs.

At any given time three or four Muons are on ridges — Forest Lands High Point, Forest Lands Ridge, Bowman Mountain — and the rest are on the bench being upgraded or repaired. Winter on a ridge is hard on batteries, and one of them is currently off the air for exactly that reason.

Muons and Borons publish to **Particle Device Cloud**, which sends a webhook to **Home Assistant** at home. Home Assistant republishes onto a local MQTT broker, and **Node-RED** normalizes each record and writes it to **TimescaleDB**. A Muon that has lost cellular coverage sends the same reading over LoRaWAN instead (§3), so one dead radio does not make a dead station. The custom camera posts its weather reading and heartbeat straight to the Home Assistant webhook — same shape, no Particle Cloud in the middle.

Diagram — Particle → Home Assistant → database



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3. LoRaWAN and The Things Network

LoRaWAN is a long-range, low-power radio protocol built for distributed environmental monitoring. In practice, what surprised me is how **promiscuous** the network is: put up a gateway and you extend the mesh. My sensors hear other people's traffic — I'm picking up meshes along Mosquito Lake Road — and vice versa.

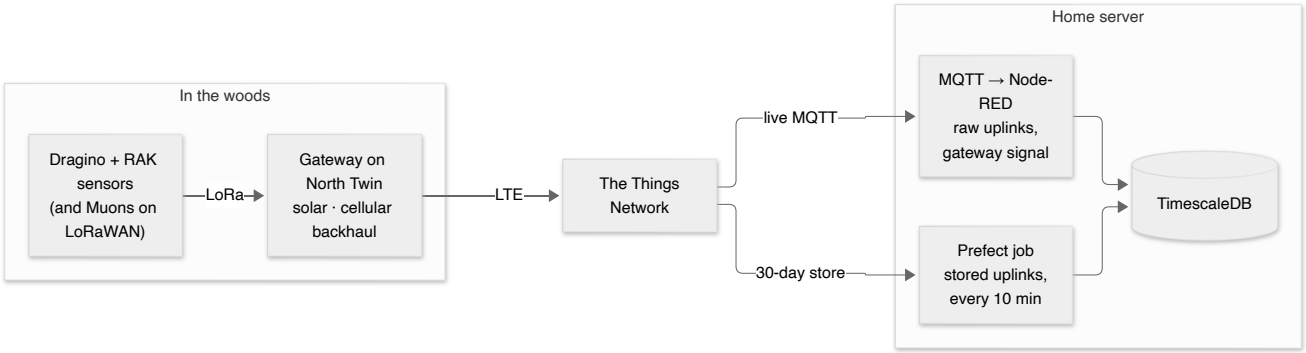
The main woods gateway sits on **North Twin** — a RAK7268V2, solar powered, **cellular backhaul**. Winter taught me that **batteries really don't like cold**: even with what I thought was enough solar and battery headroom, cold snaps shut the gateway down. Heated battery setups are the lesson from a season of alpine winter ops.

End devices are small — roughly a third of a loaf of bread, sometimes smaller. Most gear has QR-coded tags; scan one and you'll land on the project site. I guess I accidentally built **geocaching**. I hope that's fun rather than annoying.

I'm also considering gateways along **Bowman Ridge** — more accessible than North Twin, though North Twin has the elevation advantage. Still thinking that through.

Payloads go through the gateway to **The Things Network** (TTN). Two things happen at home with them. Node-RED subscribes to TTN's live MQTT feed and records every raw uplink and which gateways heard it, which is how I watch signal quality over time. Separately, a scheduled job pulls TTN's stored uplinks every ten minutes and writes the decoded weather rows — so if the house loses internet for an hour, nothing from the woods is lost; TTN keeps thirty days and the job catches up.

Diagram — LoRaWAN → gateway → TTN → database



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4. What's in the field

Most nodes fall into two families:

1. **Dragino** — rugged, solar-powered, off-the-shelf environmental sensors (temp, humidity, snow depth, soil temp, doors, buttons, etc.). Built for ranches and distributed monitoring; a good fit for the woods.
2. **Custom Particle Muons** — air quality and anything needing more compute or multiple radios.

Beyond temp and pressure, the fleet now carries **ultrasonic wind sensors**: one on the RAK weather station at Forest Lands Ridge, and one on the newest Muon build. They're not heated — winter behavior is still TBD. The hope is that keeping them around 3,500–4,000 feet, where sun cycles warm them, will prevent rime and jamming. The goal is **differential wind**: one along the skin route on the **northwest face of North Twin**, maybe one on **South Twin**, maybe one on **Bowman**, to compare what the ridges are actually doing.

A fair amount of gear came from wholesale suppliers — samples, donations, and "please sell me one unit" emails. It's not as expensive as the custom Muon sticker shock suggests, but it's still a pile of stuff to maintain in the forest.

Particle builds

Device	Role	Notes
Muon (Olivine-Wx1 … Wx6)	Weather + air quality	Temp, humidity, pressure, particulates; cellular and LoRaWAN. Field units at Forest Lands High Point, Forest Lands Ridge and Bowman Mtn. Newest build adds wind and battery telemetry over RS-485.
Boron BRN404X (two units)	Weather	First field experiment on Bowman; cellular-only proved marginal → LoRaWAN parallel path. One on the 7000 Road, one on the bench.
Raspberry Pi 5 camera (`woods-cam-003`)	Imaging + weather	The Bowman Ridge camera on the site. `woods-cam-002` is being rebuilt; the Tachyon `woods-cam-001` is retired.

LoRaWAN builds

Device	Role	Sites / notes
RAK2560 + RK900-09 + RAK9154	Full weather pack (temp, humidity, pressure, wind, battery)	Forest Lands Ridge
Dragino DDS75-LB	Snow depth	Ultrasonic distance to snow surface
Dragino S31B-LS	Temp / humidity	North Twin Approach, Baker View Turn, South Twin Approach, North Twin West Ridge, N Twin Extra, South Twin Orsino Basin, Dailey Prairie
Dragino D20S-LS	Soil temperature	Dailey Prairie ground sensor
Dragino SN50v3	Multi-sensor node	Forest Lands High Point
RAK10701 Plus	Field tester	GPS / RF diagnostics when roaming
RAK10703	Seismic	Mount Baker is an active volcano, so why not? Indoors for now.

Other fused sources

The same backend also ingests **WeatherFlow Tempest**, **PurpleAir**, **OpenWeatherMap**, and the home's own Home Assistant sensors — all into the same database (see §5).

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5. Unified backend — Postgres / TimescaleDB

Everything — cellular, LoRaWAN, trail cams, fused external sources — lands in one place:

1. **TTN** or **Particle Cloud** → MQTT at home
2. Payload decode → **common flat JSON shape**

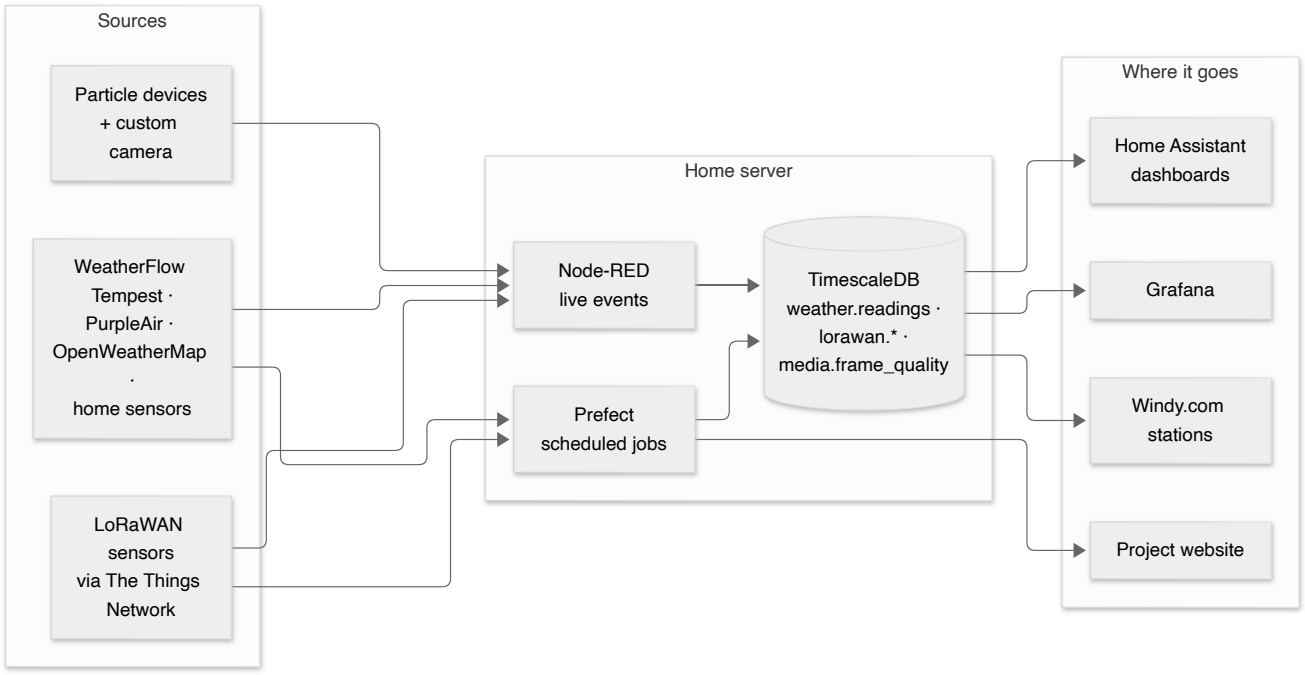
3. **Postgres / TimescaleDB** time-series storage
4. **Node-RED** for the live event paths, **Prefect** for scheduled jobs — the TTN storage pull, the camera downloads and publishing, frame-quality measurement, and the daily Twins forecast

TimescaleDB is the single time-series store.

Schema / table	Purpose
`weather.readings`	Unified hypertable; `source` distinguishes LoRaWAN, Particle, WeatherFlow, PurpleAir, etc.
`weather.power_readings`	Battery and cellular heartbeats from the Particle devices and cameras
`lorawan.*`	Raw uplinks, which gateways heard them, signal quality over time
`media.frame_quality`	Per-frame measurements for every camera image: brightness, blur, clipping, size

Grafana dashboards, **Windy** publishing, Home Assistant dashboards and the project website all read from the same database.

Diagram — field to database to you



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6. Where the data lives (and what's missing)

Most weather data you can use today is on Windy.com — Node-RED publishes the readings Windy accepts (temp, pressure, wind, etc.). Each camera on the project website links to the Windy station nearest it.

Not on Windy yet (in the database, working on public access):

- Air quality (AQI / particulate) — shown on the project site for Bowman Mtn and N Twin
- Soil and ground / probe temperatures
- Some other specialty channels

Ideas for a good public home for those streams are welcome.

Trail camera images are on the project website: **disruptivepatternmaterial.github.io/webcams**. Thousands of frames so far, seasonal timelapse in the works.

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7. Privacy, trail cameras, and being thoughtful in the woods

People go into the woods partly to **get away** from screens and surveillance. Random tech on a skin track is the last thing anyone needs. I get that.

- Cameras target **terrain and conditions**, not people.
- They have still captured a few passersby — **unintentionally**, and I'm sorry if that bothered anyone.
- **Every new frame is run through a classifier** that flags people. Right now it runs in review mode — it logs what it sees and I check its work — and it will withhold flagged frames automatically once I've verified it is accurate enough to trust. Until then, that's a manual job.
- **No faces have been posted.**
- Cameras are being **relocated** higher and farther from travel corridors where views stay useful.

If you find a QR-tagged box or a camera and have concerns, reach out via the site. I'd rather adjust placement or pull something than make the backcountry feel like a surveillance project. The intent is **safety and recreation** — conditions, smoke, wind, snow — not monitoring people.

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8. How you can help (or just say hi)

There's always a maintenance season — heated batteries, gateway moves, wind sensor placements, camera relocations — and there's a lot of gear to track in the woods.

If you're interested:

- **Hauling gear** to North Twin, South Twin, or stashing at an agreed spot
- **Field trips** — always happy to meet people who want to adventure outdoors
- **Engineering, documentation, or data ideas** — especially for the non-Windy streams

People have asked how to **chip in financially**. There's a Buy Me a Coffee link on the site for ongoing costs — solar, batteries, cellular, Starlink, parts.

If you just use the Windy stations or camera views to plan a day, that's enough. That's why I built it.